

Motor Datasheet

Operating Point: Continuous | Number of Turns: 11

Geometry

Parameter	Value	Unit
Machine Outer Diameter	50.00	mm
Machine Inner Diameter	18.00	mm

Mass Properties

Parameter	Value	Unit
Machine Mass	123.1	g
Rotor Mass	20.1	g
Stator Mass	103.0	g

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	4.012	A
D-Axis Inductance (Ld)	96.2522	uH
Q-Axis Inductance (Lq)	96.3407	uH
PM Flux Linkage	1.6946	mWb

Performance

Parameter	Value	Unit
Torque (Continuous)	72.61	mNm
BackEMF Constant (Ke)	0.8873	mV/rpm
Motor Constant (Km)	268.8302	mNm/√W
Torque Constant (Kt, no-load)	12.7433	mNm/A
Saturation	0.00	%

Losses

Parameter	Value	Unit
Total Losses	8.827	W
Resistive Losses	8.521	W

Configuration

Parameter	Value	Unit
Pole Pairs	5	-
Number of Turns	11	-